02.02.2024

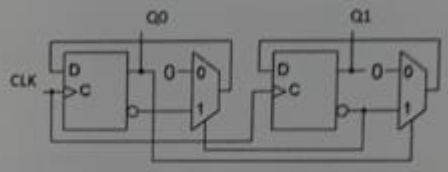
EE2003 and CSE 3017 DIGITAL DESIGN FINAL MAKE-UP EXAM

Name: ANSWERS

DURATION: 90 MINUTES
CLOSED BOOK, CLOSED NOTES EXAM. NO CALCULATORS OR CHEAT SHEETS ALLOWED.
PLEASE PROVIDE YOUR ANSWERS IN DESIGNATED PLACES.

Q1) For the sequential circuit below,

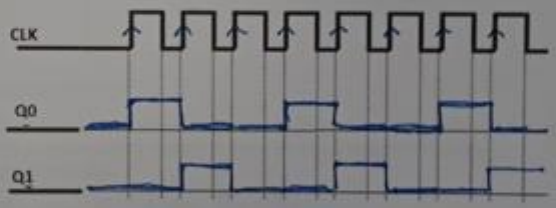
(A) **(8 points)** Write state equations.



$$Q_0[n+1] = Q_1[n]\,0 + \overline{Q_1[n]}\,\overline{Q_0[n]}$$
$$= \overline{Q_1[n]}\,\overline{Q_0[n]}$$

$$Q_1[n+1] = Q_0[n]\,0 + Q_0[n]\,\overline{Q_1[n]}$$
$$= Q_0[n]\,\overline{Q_1[n]}$$

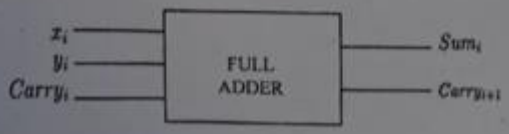
(B) **(8 points)** If the circuit starts in the state $Q_1Q_0 = 00$, sketch output states.



(C) **(4 points)** What does this circuit do?

Counts 0 - 1 - 2

Q2) A full adder circuit using has a general block diagram as below.



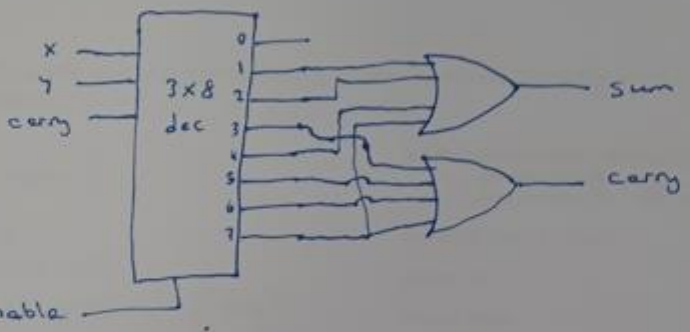
(A) **(8 points)** Form truth table. (x and y are single bit variables)

x	y	carry	sum	carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$sum = \Sigma(1,2,4,7)$

$carry = \Sigma(3,5,6,7)$

(B) **(10 points)** Implement the circuit using a 3x8 decoder and appropriate logic gates.



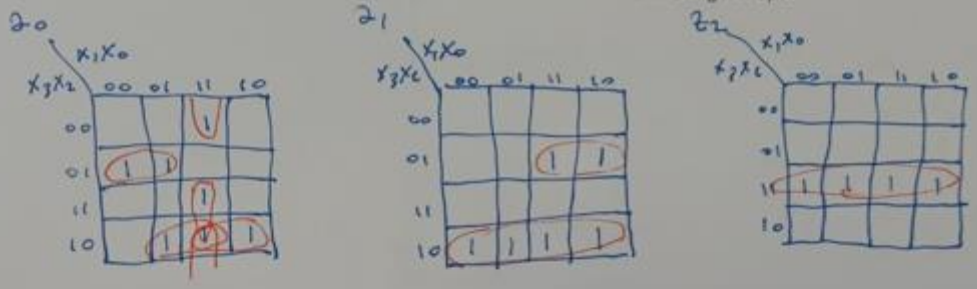
Q3) Design a combinational logic circuit which receives a 4-bit unsigned number X (x_3 x_2 x_1 x_0) as input and produces an output Z which equals the result of integer division of X by 3 (e.g., if X=7, Z=2).

(A) **(8 points)** Form the truth table.

X_3	X_2	X_1	X_0	Z_2	Z_1	Z_0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	0	1	0	0	0	0
0	0	1	1	0	0	1
0	1	0	0	0	0	1
0	1	0	1	0	0	1
0	1	1	0	0	1	0
0	1	1	1	0	1	0
1	0	0	0	0	1	0
1	0	0	1	0	1	1
1	0	1	0	0	1	1
1	0	1	1	0	1	1
1	1	0	0	1	0	0
1	1	0	1	1	0	0
1	1	1	0	1	0	0
1	1	1	1	1	0	1

2

(B) **(10 points)** Determine the minimized output functions in SoP form using K-maps.



$$z_2 = x_3 x_2$$

$$z_1 = x_3 \bar{x}_2 + \bar{x}_3 x_2 x_1$$

$$z_0 = x_1 x_0 (\bar{x}_2 + x_3) + x_3 \bar{x}_2 (x_0 + x_1) + \bar{x}_3 x_2 \bar{x}_1$$

Q4) (10 points) For a RAM application (with depth = 4 and width = 8), an excerpt of testbench file is given below. Write main module.

(24)

```
module ram (
input clock,
input write_enable,
input [3:0] address, // depth
input [7:0] data_in, // width
output [7:0] data_out );

reg [7:0] ram [15:0];

always @ (posedge clock)
    if (write_enable)
        ram [address] <= data_in;

assign data_out = ram [address]

endmodule
```

```
ram UUT(.clock(clock), .write_enable(write_enable),
        .address(address),
        .data_in(data_in), .data_out(data_out));

always #2 clock = ~clock;

integer i;

initial begin
    clock = 0;
    write_enable = 0;
    address = 4'h0;
    data_in = 8'h00;
    #9 write_enable = 1;

    for(i = 0; i < 16; i = i + 1) begin
        address = i;
        data_in = i * 16;
        #4;
    end

    write_enable = 0;

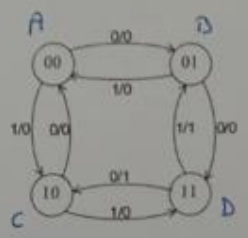
    for(i = 0; i < 16; i = i + 1) begin
        address = i;
        #4;
    end

end
```

Q5) Given the state diagram below,

(A) **(6 points)** Form state table.

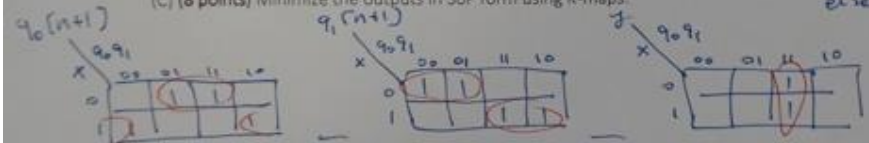
x	$q_0(n)$	$q_1(n)$	$q_0(n+1)$	$q_1(n+1)$	y
0	0	0	0	1	0
1	0	0	1	0	0
0	0	1	1	1	0
1	0	1	0	0	0
0	1	0	0	0	0
1	1	0	1	1	0
0	1	1	1	0	1
1	1	1	0	1	1



(B) **(10 points)** Write Verilog description using case statements.

```
module But(x, clk, reset, y);
input x, clk, reset;
output reg y;
reg [1:0] state = 2'b00;
parameter A=2'b00, B=2'b01, C=2'b10, D=2'b11;
always @ (posedge clk)
begin
if (reset == 1)  state <= A;
else
begin
if(x==0)
  case (state)
  A: state <= B;
  B: state <= D;
  C: state <= A;
  D: state <= C;
  endcase
if(x==1)
  case(state)
  A: state <= C;
  B: state <= A;
  C: state <= D;
  D: state <= B;
  endcase
if (((x==0)&&(state==D)) ||
    ((x==1)&&(state==D))) y<=1;
else y<=0;
end
end
endmodule
```

(C) **(8 points)** Minimize the outputs in SoP form using K-maps.



$q_0(n+1) = \bar{x}\,q_1(n) + x\,\bar{q}_1(n)$

$= x \oplus q_1(n)$

$q_1(n+1) = \bar{x}\,\bar{q}_0(n) + x\,q_0(n)$

$= \overline{x \oplus q_0(n)}$

$y = q_0(n)\,q_1(n)$

(D) **(10 points)** Implement the sequential circuit using D flip flops.

